Performance Evaluation

PPO $\times$ Macro Adaptive Strategy (Live Trading)

Owen (Presenter) Alina Jeffery Vicky Vivian

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Outline

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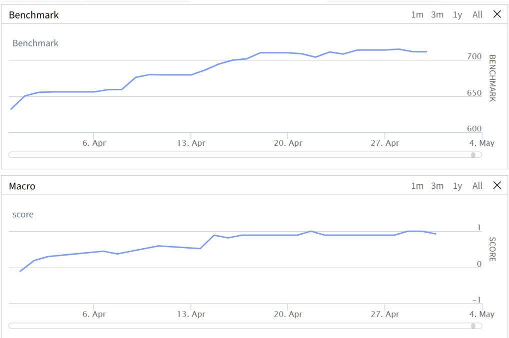
Live Trading Performance Metrics (April 2026)

Metric	Backtest	Live (Apr 1–30, 2026)	Difference
CAGR (%)	121.03	108.50	12.53
Annualised Standard Deviation (%)	6.56	7.69	-1.13
Sharpe Ratio (rf=0%)	9.5435	8.2936	1.2499
Max Drawdown (%)	0.7	0.6	0.1
Beta (vs. SPY)	0.4819	0.6534	-0.1715
Alpha (annual, %)	11.67	-5.31	16.98
Win Rate (%)	47.87	48.40	-0.53
Drawdown Recovery (days)	5	3	2
Daily Turnover (%)	33.50	23.65	9.85

Period: April 1, 2026 – April 30, 2026 (21 trading days).

Equity Curves: Strategy vs. Benchmark (April 2026)

(a) Live Strategy Equity Curve



(b) Benchmark (SPY) Equity Curve

Live period: April 1–30, 2026 (21 trading days).
Benchmark: SPY ETF, normalised to same starting value.

Outline

Cost Model Parameters (Live)

Cost Component	Parameter Value	Notes
Commission (IB Tiered)	$0.0035 / share	Minimum $0.35 per order
Slippage (one-way)	0.5 bps	SpreadSlippageModel
Borrow Cost (annual)	3.00%	Applied to short positions
Dividend WHT	30%	Only on long holdings
Lambda (fee penalty)	1.0	PPO reward function
Lambda (borrow penalty)	1.0	PPO reward function
Eta (turnover penalty)	0.05	PPO reward function
Lambda (tax penalty)	1.0	PPO reward function

Penalty coefficients encourage agent to minimise frictions.

Outline

Finding 1: Live Transaction Costs Exceed Backtest

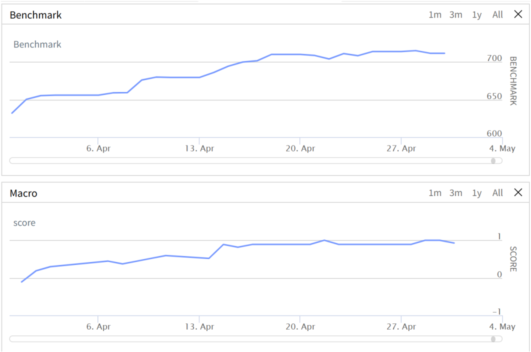
- **Problem:** Large position sizes cause market impact and higher-than-expected commissions.
- Example: two short positions entered on April 1, exited on April 2:

Date	Symbol	Shares	Notional (USD)	Commission (USD)
2026-04-01	U	9,425	206,774	46.05
2026-04-01	CDE	75,550	1,418,003	543.86

- CDE trade: \$543.86 commission (vs. backtest assumption of ~\$264 for same notional).
- **Root cause:** IB tiered pricing becomes expensive for very large share counts; also wider spreads hurt slippage.
- **Mitigation:** Re-calibrate slippage to 0.75bps; break large orders into child chunks; consider futures for macro exposure.

Finding 2: Positive Macro Score Pushed Capital to SPY

- The macro score (weighted vote of 8 indicators) **never dropped below zero** in April 2026.
- Consequence: allocator gave high weight to SPY, low weight to PPO.



April 2026 macro score evolution. No negative readings.

Finding 3: PPO Crisis Alpha Remains Unconfirmed

- PPO is designed to generate **crisis alpha** during market sell-offs (high volatility, negative trend).
- April 2026 was a **bull market** – no crisis conditions occurred.
  - VIX remained below 18.
  - Macro score never signalled bearish regime.
- Therefore, we **cannot yet verify** whether the PPO agent still delivers the promised downside protection.

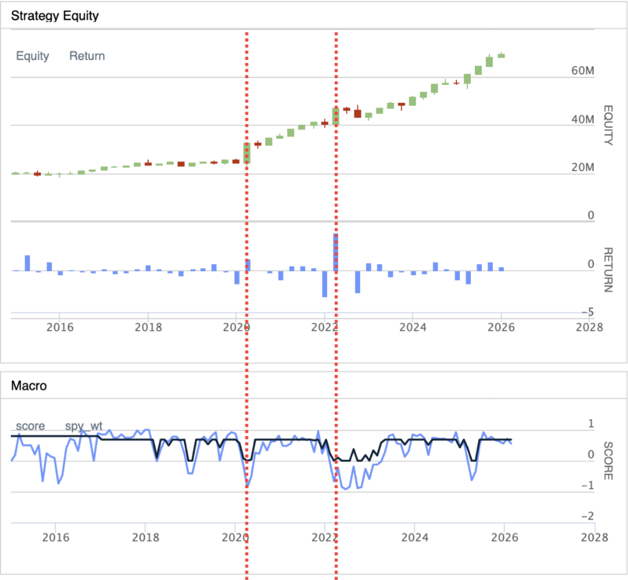
Equity Curves: Strategy vs. Benchmark (March 2026)

(a) Strategy Equity Curve (March 2026)

(b) Benchmark (SPY) Equity Curve (March 2026)

Period: March 1–31, 2026.
Benchmark: SPY ETF, normalised to same starting value.

Crisis Alpha



Trade-off: Macro Score Sensitivity

	Actual Market Regime	
	Bull Market	**Crisis (Bear)**
Signal: Crisis	False Positive (Whipsaw: PPO activated)	True Positive (Crisis alpha harvested)
Signal: Bull	True Negative (Low cost, normal returns)	False Negative (Missed crisis, large drawdown)

Trade-off:

- Too sensitive → frequent false alarms (costly whipsaw).
- Too insensitive → miss real crises (large losses).
- Our asymmetric EMA (fast ramp-up, slow decay) balances both.

Outline

Kill Switch Logic

Automatic Shutdown Conditions

1. **Drawdown trigger:** Current drawdown $>$ Historical max drawdown $\times$ 1.25
   - Historical max (backtest + live) $= 16.20\%$
   - Trigger level $= 20.25\%$
   - *Current drawdown: 0.6% (April) — no trigger*
2. **Daily loss circuit breaker:** Realised + unrealised P&L $< -10\%$ of NAV within a single day.
3. **Gross leverage exceedance:** Gross exposure $> 1.5\times$ NAV for two consecutive rebalances.

Capacity Limits & Scaling Considerations

Parameter	Current AUM	Soft / Hard Cap
Notional AUM	\$20.00M	Soft: \$50M, Hard: \$100M
Average position size	~\$240K (1.2% of AUM)	
Percent of ADV (top holdings)	<0.05%	Soft: 0.5%, Hard: 2.0%

- **Soft cap actions:** decompose orders into child chunks, switch to VWAP execution.
- **Hard cap:** stop accepting new capital; consider futures for SPY exposure to reduce stock count.
- **Monitoring:** Daily report of top 5 positions by %ADV.

Real-Time Risk Dashboards

- **Pre-trade risk guards (already active):**
  - Single position cap $<$ 20% of equity.
  - Spread alert if bid-ask $>$ 200 bps.
  - Short feasibility check before order.
- **Post-trade surveillance:**
  - End-of-day broker vs. target holdings check (tolerance 2%).
  - Daily slippage summary (average, max, 90th percentile).
  - Borrow cost accrual report.
- **Email notifications:**
  - Circuit breaker, kill switch activation.
  - Daily reconciliation mismatch $>$ 0.1% of NAV.

Outline

Key Takeaways from First Month Live Trading

- **Strategy performed as expected in a bull market:**
  - Beta rose to 0.65 (vs. backtest 0.48) – consistent with high SPY allocation.
  - Drawdown remained low (0.6%), Sharpe still healthy (8.3 annualised).
- **Three major discrepancies vs. backtest:**
  1. Live transaction costs (commissions + slippage) exceeded assumptions.
  2. Macro score never negative $\rightarrow$ PPO remained underweight; crisis alpha untested.
  3. Borrow cost and slippage data not yet fully reconciled.
- **No evidence of model failure:** The open question is whether PPO still harvests crisis alpha – requires a true market drawdown.

Immediate Action Items

Action	Deadline / Priority
Re-calibrate slippage model (0.5 $\rightarrow$ 0.75 bps)	High – May 2026
Complete borrow cost & slippage reconciliation	High – May 2026
Implement child-order chunking for large notional trades	Medium – Q2 2026
Continue live monitoring; trigger kill switch if drawdown $>$20%	Ongoing
Review PPO performance after first crisis event (VIX $>$22)	Conditional

Next formal review: End of May 2026 or after a VIX spike $>$22, whichever comes first.

Q & A

Questions & Discussion

Appendix A1: PPO Network Architecture

Policy Network $\pi_\theta(a \mid s)$: 3-layer MLP (128 hidden units each, Tanh activations). Outputs mean $\mu(s)$ and learned log-std $\log \sigma$.

$$a \sim \mathcal{N}\big(\mu_\theta(s),\ \mathrm{diag}(\sigma^2)\big), \qquad w = \frac{\tanh(a)}{\|\tanh(a)\|_1}$$

$w \in \mathbb{R}^{15}$ are L1-normalised factor weights.

Value Network $V_\phi(s)$: 3-layer MLP (128 hidden, Tanh) $\rightarrow$ scalar value estimate.

State vector ($d = 87$): 12 core features $+ 5 \times 15$ factor statistics.
Core: log(equity), cash%, gross lev, β, w_L, w_S, n_{pos}, turnover, tc%, borrow%, β_{last}, TE.
Per factor ($k = 1 \ldots 15$): $\overline{\mathrm{IC}}_k$, $\sigma_{\mathrm{IC},k}$, $t_{\mathrm{IC},k}$, $\overline{\mathrm{QS}}_k$, $\sigma_{\mathrm{QS},k}$.

Action ($d = 15$): one weight per alpha factor.

Appendix A2: PPO Objective & GAE

Clipped Surrogate Objective:

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t\Big[\min\big(r_t(\theta)\,\hat{A}_t,\;\; \text{clip}(r_t(\theta),\, 1-\epsilon,\, 1+\epsilon)\,\hat{A}_t\big)\Big], \qquad r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)},\ \epsilon = 0.2$$

Generalised Advantage Estimation ($\gamma = 0.99$, $\lambda = 0.95$):

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l\,\delta_{t+l}, \qquad \delta_t = r_t + \gamma\,V(s_{t+1}) - V(s_t)$$

Value loss: $L^V(\phi) = \frac{1}{N}\sum_t (V_\phi(s_t) - \hat{R}_t)^2, \quad \hat{R}_t = \hat{A}_t + V(s_t).$

Training: 10 epochs, batch size 256, gradient clip 1.0. Updated every 5 trading days (after ramp-up ends).

Appendix A3: Macro Score — 8 Indicators

The macro allocator computes a weighted vote:

$$s_{\text{macro}} = \frac{\sum_i v_i \cdot \omega_i}{\sum_i \omega_i}, \qquad v_i \in \{-1,\, 0,\, +1\}$$

i	**Indicator**	ω_i	**Bullish** $(+1)$	**Bearish** (-1)
1	SPY / SMA200 − 1	2.0	$> 3\%$	$< -2\%$
2	SMA50 / SMA200 − 1	1.5	$> 1\%$	$< -1\%$
3	252-d momentum	1.0	$> 5\%$	$< -5\%$
4	Drawdown from 252-d hi	2.0	$> -3\%$	$< -8\%$
5	VIX level	2.5	< 16	> 22
6	Yield 10Y−2Y	1.0	> 0.50	< 0
7	20-d momentum	2.0	$> 3\%$	$< -4\%$
8	50-d momentum	1.5	$> 4\%$	$< -5\%$

Indicators between both thresholds vote 0 (neutral). Total weight $\Sigma\omega = 13.5$; score range $\approx [-1, +1]$.

Appendix A4: Score Dynamics — Trend, Vol, Acceleration

From the score buffer (up to 320 observations):

Trend (fast mean − slow mean):

$$\text{trend} = \bar{s}_5 - \bar{s}_{20}$$

Volatility (rolling std of recent scores):

$$\text{vol} = \text{std}(s_{-10:})$$

Volatility acceleration (change in short-term vol):

$$\text{vol\_accel} = \text{std}(s_{-5:}) - \text{std}(s_{-10:-5})$$

These three quantities feed directly into the crisis intensity computation. They capture the **speed and direction** of regime changes, not just the current score level. A sharp negative trend combined with rising volatility signals a crisis is **accelerating**.

Appendix A5: Crisis Intensity — Four Signals

Four normalised crisis signals (all clamped to $[0, 1]$):

$$\sigma_{\text{trend}} = \text{clamp}\left(\frac{-\text{trend} - 0.03}{0.22},\ 0,\ 1\right), \qquad \sigma_{\text{vol}} = \text{clamp}\left(\frac{\text{vol} - 0.05}{0.20},\ 0,\ 1\right),$$

$$\sigma_{\text{accel}} = \text{clamp}\left(\frac{\text{vol\_accel} - 0.01}{0.09},\ 0,\ 1\right), \qquad \sigma_{\text{level}} = \text{clamp}\left(\frac{-s_{\text{macro}} - 0.10}{0.30},\ 0,\ 1\right).$$

Composite raw crisis:

$$c_{\text{raw}} = 0.50\ \min(\sigma_{\text{trend}}, \sigma_{\text{vol}}) + 0.25\ \sigma_{\text{vol}} \cdot \sigma_{\text{accel}} + 0.25\ \sigma_{\text{level}}$$

The primary term requires **both** negative trend and high volatility simultaneously (the minimum operator). The bonus terms reward acceleration and deeply negative macro scores.

Appendix A6: Crisis Intensity — Asymmetric Smoothing

The raw crisis score is smoothed with an **asymmetric EMA**—fast ramp-up, slow decay:

$$c_t = \alpha\, c_{\text{raw}} + (1-\alpha)\, c_{t-1}$$

$$\alpha = \begin{cases} 0.70 & c_{\text{raw}} > c_{t-1} \;\; \text{(entering crisis — react fast)} \\ 0.60 & \text{trend} > 0.05 \;\wedge\; s > -0.5 \;\; \text{(strong recovery signal)} \\ 0.40 & \text{trend} > 0 \;\; \text{(mild recovery)} \\ 0.15 & \text{otherwise (slow decay — stay defensive)} \end{cases}$$

Why asymmetric?

- Entering crisis: $\alpha = 0.70$ means 70 % of the new signal is adopted immediately. We shift capital to PPO within 1–2 days.
- Exiting crisis: $\alpha = 0.15$ means we wait $\sim$6–7 days to confirm recovery before shifting back to SPY. Avoids whipsaw from dead-cat bounces.

Appendix A7: SPY Weight — C-Style Dynamic Map

We use only the C-style map (`dynamic_w` $= 1$, `static_w` $= 0$):

$$w_c(s) = \mathrm{clamp}\Big(\underbrace{0.7}_{\text{base}} + \frac{s}{\hat{\sigma}_{\text{long}}},\ 0,\ 1\Big), \qquad \hat{\sigma}_{\text{long}} = \mathrm{std}\big(s_{\text{buf}}[-260:]\big)$$

Intuition: when the macro score is positive and large relative to its historical variability, SPY gets high weight. When the score is negative, SPY weight drops toward zero—capital flows to PPO.

Crisis-adjusted SPY target:

$$w_{\text{spy}} = w_c \times (1 - c_t)$$

Floor constraint: PPO always gets at least 30 % gross.

$$w_{\text{spy}} \le 1 - \frac{g_{\min}}{0.95} = 1 - \frac{0.30}{0.95} \approx 0.684$$

Appendix A8: PPO Gross Exposure and Capital

Given the SPY weight, the remaining capital goes to PPO:

PPO gross exposure:

$$g_{\text{ppo}} = 0.95\,(1 - w_{\text{spy}})$$

PPO capital:

$$K_{\text{ppo}} = (1 - w_{\text{spy}}) \times \text{Equity}$$

Worked example:

$$s_{\text{macro}} = 0.4, \quad c_t = 0.1, \quad \hat{\sigma}_{\text{long}} = 0.25, \quad \text{Equity} = \$20\text{M}$$

Step	Computation
C-style base	$w_c = \text{clamp}(0.7 + 0.4/0.25,\ 0,\ 1) = \text{clamp}(2.3) = 1.0$
Crisis adjust	$w_{\text{spy}} = 1.0 \times (1 - 0.1) = 0.90$
Cap enforce	$w_{\text{spy}} = \min(0.90,\ 0.684) = 0.684$
PPO gross	$g_{\text{ppo}} = 0.95 \times (1 - 0.684) = 0.300$
PPO capital	$K_{\text{ppo}} = 0.316 \times \$20\text{M} = \$6.32\text{M}$

Appendix A9: Full Allocation Pipeline

8 Macro Indicators → Weighted Vote → Score Buffer → Trend, Vol, Vol Accel → Crisis Intensity c_t → $w_{\text{spy}},\ g_{\text{ppo}}$

Summary of key formulas:

$s_{\text{macro}} = \dfrac{\sum v_i \omega_i}{\sum \omega_i}$	(weighted vote of 8 indicators)
$c_t = \alpha\, c_{\text{raw}} + (1-\alpha)\, c_{t-1}$	(asymmetric EMA, $\alpha \in \{0.15, 0.40, 0.60, 0.70\}$)
$w_{\text{spy}} = \text{clamp}\big(0.7 + s/\hat{\sigma},\ 0,\ 1\big) \times (1 - c_t)$	(C-style $\times$ crisis discount)
$g_{\text{ppo}} = 0.95\,(1 - w_{\text{spy}})$	(residual to PPO)

Appendix A10: Ramp-Up Period Allocation

During the first 2 years (2015-01-01 to 2017-01-01), allocation is **fixed** regardless of macro conditions:

$$w_{\text{spy}} = 0.80, \qquad g_{\text{ppo}} = 0.20$$

PPO capital during ramp-up:

$$K_{\text{ppo}} = \text{Equity} \times \frac{g_{\text{ppo}}}{\text{gross\_budget\_frac}} = \text{Equity} \times \frac{0.20}{0.95} \approx 0.211 \times \text{Equity}$$

Why this design?

- PPO's random policy creates unpredictable L/S positions.
- 80 % SPY provides a stable capital base, preventing margin calls.
- Even a total wipeout of the PPO book costs only ∼10 % of portfolio.
- Macro dynamics (trend, vol, crisis) are still tracked internally, so the allocator is "warm" when ramp-up ends.

Appendix A11: 15 Alpha Factors

k	**Factor**	**Formula (sign = desired direction)**
1	mom_5	$P_t/P_{t-5} - 1$
2	mom_20	$P_t/P_{t-20} - 1$
3	mom_60	$P_t/P_{t-60} - 1$
4	rev_2	$-(P_t/P_{t-2} - 1)$
5	rev_5	$-(P_t/P_{t-5} - 1)$
6	vol_10	$-\mathrm{std}(r_{t-9:t})$
7	vol_20	$-\mathrm{std}(r_{t-19:t})$
8	hlv_20	$-\mathrm{mean}\big((H-L)/P\big)_{20}$
9	rsi_14	$-\mathrm{RSI}(14)$
10	trend_20	$P_t/\mathrm{MA}_{20} - 1$
11	gap_1	$P_t/P_{t-1} - 1$
12	skew_20	skewness of r_{20}
13	kurt_20	excess kurtosis of r_{20}
14	dvol_20	$-\mathrm{std}(r_{<0})_{20}$
15	maxdd_60	max drawdown over 60 d

All factors are cross-sectionally winsorised (1 %) then z-scored. PPO outputs $w \in \mathbb{R}^{15}$; stock score $= Z_i^\top w$.

Appendix A12: Portfolio Construction & Beta Hedge

Scoring: $\text{score}_i = Z_i^\top w, \quad Z \in \mathbb{R}^{N\times 15}, \;\; w \in \mathbb{R}^{15}$ (PPO action).
Selection: Top-80 by score $\rightarrow$ long; Bottom-80 $\rightarrow$ short.
Weighting: Within each side, weight $\propto |\text{score}|$; rescaled so $\sum |h_L| = \frac{0.95}{2} K_{\text{ppo}}$ (same for short side).
Position bounds: $h_{\min} = 0.5\,\% \cdot K_{\text{ppo}}, \;\; h_{\max} = 15\,\% \cdot K_{\text{ppo}}$.

Beta neutralisation (up to 40 iterations):

1. Compute $\hat{\beta} = \sum_i h_i \beta_i / K_{\text{ppo}}$.
2. If $|\hat{\beta}| \leq 0.10$, stop.
3. Find best (i, j) pair (one long, one short) whose $\beta_i - \beta_j$ can tilt $\hat{\beta}$ toward 0.
4. Shift Δ dollars from j to i subject to position bounds.
5. Repeat.

Appendix A13: PPO Reward Function

The agent receives a **PPO-isolated return** (SPY contribution stripped):

$$r_{\text{ppo}} = \frac{r_{\text{portfolio}} - w_{\text{spy}} \cdot r_{\text{SPY}}}{g_{\text{ppo}}}$$

Composite reward:

$$R = r_{\text{ppo}} - \underbrace{\lambda_\beta \ \max(0, \ |\beta| - 0.10)}_{\text{beta penalty}} - \underbrace{\lambda_f \cdot \text{fee\%}}_{\text{commission}} - \underbrace{\lambda_b \cdot \text{borrow\%}}_{\text{borrow cost}} - \underbrace{\eta \cdot \text{turnover\%}}_{\text{turnover}} - \underbrace{\lambda_\tau \cdot \text{tax\%}}_{\text{div WHT}}$$

Coefficient	Value
λ_β	2.0
λ_f	1.0
λ_b	1.0
η	0.05
λ_τ	1.0

Appendix A14: Transaction Cost Formulas

Slippage (`SpreadSlippageModel`):

$$\text{slip} = P \times 0.00005 \quad \text{per side (0.5 bps)}$$

Commission (`IBTieredFeeModel`):

$$\text{fee} = \max\bigl(\$0.35,\ \ \min(Q \times \$0.0035,\ \ Q \cdot P \cdot 0.005)\bigr)$$

Daily borrow charge:

$$\text{borrow}_t = \sum_{i:\, h_i<0} |h_i \cdot P_i| \times \frac{0.03}{252}$$

Dividend withholding tax:

$$\text{WHT}_j = D_j \times Q_j \times 0.30 \qquad (Q_j > 0 \text{ only})$$

Total execution cost per period:

$$C = \sum \text{slippage} + \sum \text{fee} + \sum \text{borrow}_t + \sum \text{WHT}_j$$